# SECTION 16230-A
# HIGH SPEED ENGINE - GENERATORS

## PART 1 - GENERAL

### 1.1 DESCRIPTION

This specification covers designing, furnishing, installing, and commissioning the 480V diesel engine generator sets complete with paralleling, and synchronizing.

### 1.2 APPLICABLE CODES AND STANDARDS

The standards and codes applicable to only a portion of the work specified in this specification are referenced in the relevant parts and clauses. Codes and standards which are generally applicable to the work of this specification are listed below:

| Sponsor | Number | Title |
|---|---|---|
| ANSI | B110.1 | Steel Inside Tanks for Oil-Burner Fuel |
| ANSI | C37.90 | Relays and Relay Systems Associated with Electric Power Apparatus |
| ANSI | C39.1 | Electrical Analog Indicating Instruments |
| ANSI | C50.1 | Synchronous Generators, Motors and Machines in General |
| ANSI | C50.12 | Requirements for Salient Pole Generators and Condensers |
| ASME | Division 1 | Boiler and Pressure Vessel Code, Pressure Vessels |
| ASME | PTC-16 | Speed Governing Systems for Internal Combustion Engine-Generator Units |
| ASME | PTC-17 | Performance Test Codes for |

| | | |
|---|---|---|
| | | Internal Combustion Engines |
| ASTM | B 21 | Naval Brass Rod, Bar and Shapes |
| ASTM | B 111 | Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock |
| DEMA | | Recommended Practices for stationary Diesel and Gas Engines |
| IEEE | 85 | Test Procedures for Airborne Sound Measurements on Rotating Electric Machinery |
| IEEE | 115 | Test Procedure for Synchronous Machines |
| IEEE | 126 | Speed Governing Internal Combustion Engine-Generator Units |
| IEEE | 446 | Recommended Practice for Emergency and Standby Power System |
| NEMA | ICS 1 | Industrial Controls and Systems |
| NEMA | MG 1 | Motors and Generators |
| NEMA | MG 2 | Safety Standards for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators |
| NEMA | E1-1 | Electrical Indicating Instruments |
| NEMA | MG 3 | Sound Level Prediction for Installed Rotating Electric Machines |
| NFPA | 30 | Flammable and Combustible Liquid Code |

| | | |
|---|---|---|
| NFPA | 37 | Installation and Use of Stationary Combustion Engines |
| NFPA | 70 | National Electrical Code |
| TEMA | | Standards for Tubular Exchanger Manufacturers Association |
| UL | 80 | Steel Above Ground Tanks Oil-Burner Fuel |
| UL | 142 | Steel Above Ground Tanks for Flammable and Combustible Liquids |
| UL | 2200 | Stationary Engine Generator Assemblies |

## 1.3 SUBMITTALS

Submit the following for all equipment components and accessories:

A. Single line diagrams

B. Schedule including the fabrication, factory testing, delivery and field installation.

C. Schematic and interconnection diagrams

1. Schematic diagrams of the following:

   a. Engine control.

   b. Generator control.

   c. Automatic engine starting.

2. Interconnection diagrams showing all electrical and piping connections between items of equipment.

D. Generator and control panel wiring diagrams and outline drawings.

E. Unit outline drawings.

F. Detail drawings and/or shop drawings.

G. Assembly and installation drawings and manuals.

H. Parts and special tool lists.

I. Performance data, including curve of engine fuel consumption verses KW output.

J. Operations and maintenance manuals.

K. Certificate of compliance.

L. Test reports.

M. Description of the system operation.

N. Ventilation and cooling water requirements.

## 1.4 SCOPE OF INSTALLATION

The installation of the generator system shall include the following:

A. Engine-driven generator sets.

B. Paralleling and synchronizing panels and switchgears.

C. Control system.

D. Silencers, sound attenuator and chimneys.

E. Fuel supply system (Piping between ball valve that outside of day tanks room and generator sets).

F. Generator set accessories.

G. Mounting system.

H. Cast resin encapsulated dry transformers with enclosures.

I. Grounding resistors with enclosures.

J. Busway feeder work between generators and transformers.

K. Cables and cable trays between transformers and grounding resistors.

L. Generator start battery and battery charger.

M. DC charger and battery for control.

## 1.5 RELATED WORK INCLUDED

A. Vibration isolation.

B. Plumbing piping.

C. Control wiring between all automatic transfer switches and generator control system.

D. Fuel / water separators.

E. Fuel supply for the system commissioning.

F. The protocol (including that for SRS according to IEC 61850) of PLCs shall be provided to the SCADA contractor.

G. Sound attenuator for air outlet.

H. Mechanical ventilation system shall be included if the bidder considers the existing ventilation system is inadequate.

I. Provide the required documents for obtaining the Owner approval.

## 1.6 RELATED WORK EXCLUDED

A. Building shelf.

B. Ventilation system.

C. Diesel storage tanks.

D. Day tanks.

E. Reinforced concrete foundation.

## 1.7 PERSONNEL TRAINING

A. The Bidder shall provide an outline of his proposed training program. The outline of the program shall describe:

1. Type of training offered.

2. Content of the training course.

3. Method of instruction.

4. Job title and qualification of the instructors.

5. Any other information which will add to the description of the program.

B. The following items shall be the minimum quantity of tools which contractor shall provide.

| | TRAINING COURSE | NUMBER | DURATION(HOURS) |
|---|---|---|---|
| • | SYSTEM INTRODUCTION | 5(T) | 2 |
| • | OPERATION PROCEDURE AND ROUTINE MAINTENANCE | 5(T) | 6 |
| • | FAULT FINDING AND TROUBLESHOOTING | 5(T) | 6 |

(T):
MAINTENANCE TECHNICIANS

C. Annual personnel training of at least one(1) day shall be provided as requested by the Owner.

## PART 2 - SYSTEM DESCRIPTION

### 2.1 GENERAL

The system shall consist of six(6) generator sets initially and twelve(12) generator sets finally and shall include all controls, protection, wiring and accessories for unattended operation.

Each generator shall have a site capability of 2000KW, 2500KVA, 0.8 power factor, 480 volts, wye connected, three phase, four wire, 60 hertz, 1800 rpm. This power shall be applied for stand-by. The control system shall be designed to extend without modification for the paralleling of 12 generators total.

### 2.2 SYSTEM FUNCTION

A. General: The generator set shall include the capability of automatically controlling generator set operation. After starting, the unit will attain rated speed and voltage, and accept rated load. Generator set speed shall be controlled by the engine governor, while generator output voltage regulation shall be a function of the generator automatic voltage regulator. Manual adjustment of generator speed and voltage shall be provided.

1. The generator sets will automatically start, attain rated speed and voltage, and accept and share load. Starting and stopping sequence can be initiated manually or automatically by closing and opening of a contact.

2. The generators will be initiated automatically by the following methods:

   a. When automatic transfer switch senses the normal power failure, all generators will start automatically and be synchronized.

3. When considering the severe response of one (1) generator supplying the inrush current to all step-down transformers, the generator control system will close the distribution circuit breakers sequentially to keep the generator from shut-down. Only if the contractor considers it is necessary, the function is required.

4. If required, the distribution circuit breakers shall be used to establish the priority of the emergency loads.

5. When normal power restores, the automatic transfer switches will signal the generators and the generators will begin to cool down after receiving the signal from all automatic transfer switches and after the pre-determined delay time.

6. The generator control system will be able to communicate with the Power SCADA system to allow the Power SCADA system to add or remove the loads according to the actual capacity of active generators.

7. The generator set shall immediately shut down in the event of over-speed and low oil pressure. The high water temperature shall alarm first, then shut down the generators when reaching at higher water temperature. Cause of shutdown shall be indicated by a light annunciation and audio alarm. System logic shall prevent restart until fault is cleared.

8. The set shall have a provision for manual shutdown in

the event of an emergency.

9. Each generator set shall be capable of paralleling manually and automatically to the system bus. Appropriate monitors and safety devices will be included to provide system protection while paralleling and operating on the system bus.

10. automatic operation shall include control devices to compare and adjust oncoming units to frequency and phase of system bus, sense matching within acceptable limits, and signal closure of the circuit breaker. Control circuits shall prevent more than one circuit breaker from simultaneously closing onto a dead bus.

11. Under-frequency sensing of system bus shall be an indication of overload. Load shed circuits will communicate with Power SCADA system to initiate removal of selected load from system bus.

B. Power Rationing and Peak Shaving Operation:

In addition to emergency standby operation, the system shall be capable of power rationing and peak shaving operation. Generator sets shall be automatically / manually paralleled to the utility bus during this operation.

C. Paralleling with the Normal Power: The system shall be designed to be paralleling with the normal power.

PART 3 - PRODUCTS

3.1 OPERATING ENVIRONMENT

A. General: The operating environment of generator system shall be:

| | |
|---|---|
| • Altitude | less than 1000m |
| • Outside air temperature | 35 degree C |
| • Radiator supply air temperature | 40 degree C |
| • Minimum outside air temperature | 0 degree C |
| • Fuel type | No.2 diesel |

3.2 DEFINE CONDITIONS OF PERFORMANCE

A. Rating: Engine brake horsepower shall be sufficient to deliver full rated generator set 2000KW/2500KVA of standby power rating when operated at rated 1800rpm.

B. Conditions: The rating shall be based on SAE J1349 conditions of 100KPa and 25℃. The rating shall also apply at ISO 3046/1, or DIN 6271, or BS 5514 standard conditions.

C. Fuel: Diesel engines shall be able to deliver rated power when operating on NO.2 diesel fuel having 35 degree API (16℃) specific gravity.

D. Starting Time And Load Acceptance: Engines shall start, achieve rated voltage and frequency, and be capable of accepting load within ten seconds when properly equipped and maintained.

E. Block load Acceptance: Transient response shall conform to ISO 8528 of BS 5514 requirements.

F. Motor Starting: Motor starting inrush of SKVA shall result in maximum instantaneous voltage dip not greater than 30% as measured by light beam oscillograph and the voltage recovery to 93% of rated voltage shall be within 1 second.

## 3.3 ENGINE

A. General: The engine shall be a stationary, liquid cooled, four-cycle design, vertical in-line or V-type. It shall have 16 cylinders and minimum displacement of 60 liters. Use an environmentally friendly engine with at least Tier 2 or above.

B. Structure/Metallurgy

1. The engine shall be equipped with air filters, fuel filters, lubricating oil cooler, filters, and pressure gauge, water pump and temperature gauge, service hour meter, flywheel, and flywheel housing.

2. The design of the basic engine shall provide for maximum structural integrity to extend service life. Materials used in the engine shall incorporate the highest level of proven metallurgical and manufacturing technology.

C. Lubricating System: The lubrication oil pump shall be a positive displacement type that is integral with the engine and gear driven from the engine gear train. The system shall incorporate full flow filtration with bypass valve to continue lubrication in the event of filter clogging.

The bypass valve must be integral with the engine filter base or receptacle. Systems where bypass valves are located in the replaceable oil filter are not acceptable. Pistons shall be oil cooled by continuous jet spray to the underside of the crown and piston pin.

D. Diesel Fuel System: The fuel system shall be integral with the engine. It shall consist of fuel filter, transfer pump, electronic, governor. The transfer pump shall deliver fuel under low pressure to individual injector-one for each cylinder. The injector shall be controlled by electronic control module, both start and end points of injection can be varied. It fluctuates, based on engine load and speed. Electronics recognize the actual condition to determine the precise moment.

Flexible fuel line between engine and fuel supply shall be installed to isolate vibration.

The fuel transfer pump, injection pumps, rack and pinion assembly, and time mechanism shall be maintenance and adjustment free for the life of the equipment. The fuel filter shall not require changing more frequently than once per year or every 250 hours, which ever comes first.

E. Governor:

1. The engine governor shall control engine speed and transient load response within commercial and ISO 8528 or BS 5514 tolerances. It will be selected, installed, and tested by the generator set manufacturer.

2. The engine governor shall be an electronic load sharing and speed control device. Speed droop shall be externally adjustable from 0 (isochronous) to 10% from no load to full rated load.

   Steady state frequency regulation shall be within +/-0.25 percent. It shall be capable of sharing load with others when paralleled with similarly equipped engines. Speed shall be sensed by a magnetic pickup off the engine flywheel ring gear. A provision for remote speed adjustment shall be included. The governor shall incorporate provisions for limiting fuel during start-up, and include capability for actuator compensation adjustment. Protection from voltage spikes and reverse polarity shall be included.

F. Cooling System:

1. The engine jacket water cooling system shall be a closed circuit design with provision for filling, expansion, and draining. The cooling pump shall be driven by the engine. Coolant temperature shall be internally regulated to disconnect external cooling systems until operating temperature is achieved.

2. The wet side of water cooling system shall be painted with special coating, such as coupon blue, to prevent the corrosion. The painting shall be approved by the Owner.

3. Engine jacket water heat shall be discharged to the atmosphere through a close coupled radiator. The radiator shall be sized to cool the engine continuously while operating at full rated load and at site condition. The generator set supplier shall provide jacket water pump performance curves to equate flow against external resistance.

G. Inlet Air System:

1. The engine shall not require more than 0.07 cu m/min of air per brake horsepower for combustion. The air cleaner shall be mounted with dry element requiring replacement no more frequently than 250 operating hours or once each year.

2. Only single stage turbo-charging shall be allowed.

3. Aftercooler core air surfaces shall be coated with a corrosion inhibitor to minimize oxidation.

H. Exhaust System:

1. The engine exhaust system shall be installed to discharge combustion gases quickly and with minimum restriction. Heavy walled piping such as schedule 40 is preferred, with radii of 90°bends at least 1-1/2 times the pipe diameter. Piping shall be installed with 229mm minimum clearance from combustible material or incorporate appropriate insulation and shielding. Piping shall be supported and braced to prevent weight or thermal growth from being transferred to the engine and flexible expansion fittings provided to accommodate thermal growth.

Support dampers and springs shall be included where necessary to isolate vibration. Long runs of pipe shall be pitched away from the engine and water traps installed at the lowest point. Exhaust stacks shall be extended to avoid nuisance fumes and odors, and outlets cut at 45°to minimize noise. The exhaust pipe outlet shall be installed the anti-insect screen.

2. The exhaust silencer shall be sized and supplied by the engine supplier. It shall be mounted near the engine to minimize noise and condensation, and pitched away from the engine. A provision for draining moisture shall be included.

3. The silencer shall be critical(residential) grade type and suitable noise control for standby system.

4. The silencer shall have an end inlet and end outlet.

I. Wiring And Conduit: Engine and generator control wiring shall be multi-strand, plastic insulated cable resistant to heat, abrasion, oil, water, antifreeze, and diesel fuel. Each cable shall be heat stamped throughout the entire length to identify the cable's origin and termination. Cable shall be enclosed in nylon flexible conduit which is slotted to allow for easy access and moisture to escape. Reusable bulkhead fittings will attach the conduit to generator set mounted junction boxes.

J. Starting System:

The engine starting system shall include 24 volt DC starting motor, starter relay, and automatic reset circuit breaker to protect against butt engagement. Batteries shall be maintenance free, lead acid type mounted near the starting motor. A corrosion resistant or coated steel battery rack shall be provided for mounting. Required cables will be furnished and sized to satisfy circuit requirements. The system shall be capable of starting a properly equipped engine within 10 seconds at ambient temperatures greater than 22℃.

K. Jacket Water Heater: Jacket water heaters shall be provided to maintain coolant temperature of 32℃ while the engine is idle. Heater shall accept 120 volt AC single phase power and include adjustable thermostats.

3.4 GENERATOR

A. General:

1. The generator shall be rated for stand-by service at 2000KW, 2500KVA, 0.8PF, 480 volts, three phase, four wire, 60Hz, 1800rpm.

2. The generator shall be capable of withstanding a three phase load of 300% rated current for 3 seconds, and sustaining 150% of continuous load current for 2 minutes with field set for normal rated load excitation.

B. Structure

1. The generator shall be close-coupled, single bearing, salient pole, revolving field, synchronous type with amortisseur windings in the pole faces of the rotating field.

2. The generator housing shall be one piece and mount directly to the engine flywheel housing without bolted adapters. The rotor shaft shall pilot directly into the engine flywheel and engine torque transmitted through flexible steel plates. The generator ventilating fan shall mount to the engine flywheel and act as a pressure plate to secure the flexible plates.

C. Windings:

1. Class 200 magnet wire shall be used for rotor and stator windings. All winding insulation materials shall be Class H in accordance with BS and NEMA standards. No materials shall be impervious to oil, dirt, and fumes encountered in diesel engine operating environments.

2. The revolving field coils shall be precision wet layer wound with epoxy based material applied to each layer of magnet wire. Stator shall incorporate form wound coils wrapped with multiple layers of taping material and inserted into the stator slots. The entire stator assembly shall be vacuum pressure impregnated in a Class F thixotropic polyester varnish. Windings shall be tested at two times rated voltage plus 1000 volts AC.

D. Operating Environment:

1. The generator shall be designed to operate in a

sheltered drip-proof environment.

2. The generator shall be designed for resistance to salt and moisture laden air to inhibit rusting of internal and external metal parts and the breakdown of winding insulation.

3. Generator shall be equipped with 120 volts AC single phase space heaters to minimize condensation while the generator set is idle. The heaters shall be capable of easily mounting in the assembled generator.

E. Excitation

1. The generator exciter shall be brushless with the circuit consisting of a three-phase armature and a three-phase full wave bridge rectifier mounted on the rotor shaft. Surge suppressors shall be included to protect the rotating diodes from voltage spikes.

2. The permanent magnet excitation system shall derive excitation current from a pilot exciter mounted on the rotor shaft. It shall enable the generator to sustain 300% of rated current for ten seconds during a fault condition.

F. Voltage Regulator

1. The automatic voltage regulator shall maintain generator output voltage by controlling the current applied to the exciter field of the generator.

2. The regulator shall be a totally solid state design which includes electronic voltage buildup and overcurrent protection. It shall incorporate volts per hertz characteristics with the regulated voltage a linear function proportional to frequency over a 30 to 70Hz range.

3. The regulator shall be suitable for mounting within or external to the generator assembly, and have provision for remote voltage level control.

4. The automatic voltage regulator shall be manufactured by the manufacturer of the engine-generator set. The regulator shall sense line-to-line three phases of generator output voltage and exhibit the following characteristics:

   a. Generator output voltage maintained within +/-1%

of rated value for any load variation between no load and full load.

b. Generator output voltage drift no more than +/- 0.5% of rated value at constant temperature.

c. Generator output voltage drift no more than +/-1% of rated value over ambient temperature range of -40℃ to 70℃.

d. Telephone Influence Factor (TIF) of less than 50.

e. Electronic Magnetic Interference/Radio Frequency Interference (EMI/RFI) suppressed to commercial standards.

5. The regulator shall include the following features:

   a. Voltage level rheostat shall provide generator output voltage adjustment of - 10 to 10% of nominal.

   b. At full throttle engine starting, output voltage shall not overshoot more than 5% of its rated value, with respect to the volts/Hz curve.

6. Protection shall be provided against loss of voltage sensing and long term overcurrent conditions. The overcurrent protection function shall be automatically reset when the regulator is de-energized. The regulator shall not be damaged or result in unsafe operation when subjected to open or shorted input due to sensing loss, or a short to ground or adjacent conductor.

7. The regulator module shall be sealed in a waterproof and airproof shock resistant plastic housing and shall withstand:

   a. Temperature between -20 to 70℃.

   b. Vibration of 5G'S (peak) between frequencies of 18 to 2000 Hz in three perpendicular planes, and mechanical shock of 20G'S in all three planes.

   c. Salt spray as described by MIL-STD-810C, method 509.1 and ASTM-B117.

G. Paralleling: The regulator shall include a reactive droop network to allow paralleling with other generators.

The network shall consist of current transformer, rheostat, and control circuit which shall provide 8% minimum droop at full load and 0.8 PF.

3.5 Mounting

A. General

1. The engine and generator shall be assembled to a common base by the engine-generator manufacturer to resist deflection, maintain alignment, and minimize resonant linear vibration.

2. Steel spring isolator shall be installed between the generator set base and the mounting surface. The isolators shall bolt to the base, and have a waffled or ribbed pad on their bottom surface. The pads shall be resistant to heat and age, and impervious to oil, water, antifreeze, diesel fuel, and cleaning compounds.

3.6 Control Circuit Switchgear

A. General: The controls, protection, and monitoring systems of the generator set and its operation shall be the responsibility of the generator set manufacturer. All subsystem components, interfaces, and logic shall be compatible with engine mounted devices.

The switchgear shall be completely enclosed, dead, front, indoor type (NEMA 1), suitable for self - supported floor standing. Safety to operating and maintenance personnel shall be of first importance in the design, equipment selection, and manufacture of the equipment. Arrangement of subassemblies within the panel shall permit maximum accessibility to all parts, bus, and incoming and outgoing cables.

Sheet metal members shall be bolted or welded together to achieve adequate strength and rigidity to withstand short circuit fault currents of 25,000 amperes symmetrical. Natural ventilation shall maintain a maximum enclosure temperature of 60℃ over 40℃ ambient temperature when maximum equipment is installed (including cable and bus). Bolt holes shall be provided in the floor standing panel for holddown and leveling requirements, while all enclosures will provide lifting eyes which can be removed from blind taper holes. Louvers shall be provided at top and bottom of floor standing enclosures, and shall be rain-proof and screened to minimized moisture and rodent/insect intrusion.

Doors shall not sag when all options are installed. Front covers on all instrument compartments shall be hinged and secured with latch-type hardware. Breaker compartments extending the width of the enclosure shall hinge on one side and screw secured on the opposite side. The circuit breaker compartment and cable area shall be isolated by either fixed or removable metal barriers from the instrument compartments. Physical contact with the circuit breaker shall not be possible when only the instrument door is open. Adequate wire-way for top or bottom cable entrance and/or exit shall be provided within the enclosure. Conductor sizing shall be 75℃ conductor insulation rating.

A print pocket shall be provided in each enclosure.

The sheet metal assembly shall be primed with a lacquer type surface primer to insure good bonding qualities to the metal surface and be compatible with the lacquer finish coat. The color shall be decided by the Engineer. The coat will be the finished color for all inside and outside surfaces of the sheet metal assembly.

B. CONTROL: The control system switchgear shall automatically program the engine generator sets to automatically start, automatically parallel, and automatically share system load. For automatic operation, a contact closure shall initiate engine generator set starting with automatic closing of the generator circuit breakers occurring when conditions for paralleling are within acceptable tolerances and controlled by synchronizing circuits providing automatic speed correction signals to the engine governor systems. Control circuits shall prevent more than one generator set circuit breaker from simultaneously closing onto a dead bus. Automatic load sharing shall occur after closure of the generator circuit breakers and shall be a function of the engine governor system. Voltage regulation shall be a function of the generator automatic voltage regulator systems. Opening of the initiating starting contact after the predetermined time of confirmation shall cause the generator breakers to trip open and the generator sets to run unloaded for a cooldown period and then automatically shutdown.

The control system shall allow random paralleling of twelve(12) generators. It shall permit three generators to start simultaneously and let the first set to achieve nominal voltage and frequency connect to the dead bus to make emergency power available within 10 seconds.

The initiating signal for automatic programming of the generator sets shall be provided by the station automatic transfer switches. Output signals shall be provided when the first and each sequenced engine generator set is synchronized into the emergency bus.

For Automatic operation, a contact closure shall initiate engine generator set starting. Engine starting logic shall incorporate a fail-to-start device for automatically disconnecting the starter motor in the event the engine fail to start within a pre-determined time so as to avoid undue discharge of the batteries. The logic should allow three or more cranking attempt consisting of a 5-sec crank and 5 sec, Rest cycle. In the event that the engine fails to start after this cycle, a fail-to-start visual and audio alarm should be initiated. The cranking circuit is then isolated and no further attempt at starting should be allowed until nominally reset. All status and records shall send to SCADA.

Should an engine generator set fail to start, parallel, or develop a critical running monitored fault, the control system shall cause the engine to automatically shutdown and trip its circuit breaker.

To prevent shock loading of the generator when adding or removal of a generator. Load Ramp circuitry should be provided to allow soft loading and unloading of the generator. Soft loading control should allow the generator to take up or drop load gradually.

If the load is less than a set percentage of the total load for a period of adjustable time.(the percentage is adjustable) one generator set will shut down automatically. This set will restart automatically when the load demand again exceeds a certain percentage of the total load (typically %). When normal power is restored, engines shut down in the following sequence: When the normal power recovers and sustains for a predetermined period of time, the loads shall be transferred to the normal power system smoothly. Generator's circuit breaker to trip after time delay. Engine goes into the recooling mode and stops after required time. If during this time a failure occurs again, engines will supply power as in the above sequence.

The control system shall permit manual starting, stopping, and paralleling of the generator sets. Manual control of the individual generator set's speed and voltage shall also be provided. The automatic synchronizing and load sharing circuits shall remain operative in the manual mode. Each generator set shall be manually controlled by the station operator placing the selected engine's made switch in its manual position with the generator set starting and attaining operating speed/voltage but without automatic closing of the generator circuit breaker. To close the breaker, the operator shall be required to place the manual synchronizing switch to its on position which shall activate a station synchroscope; and then turn the breaker control switch to its close position.

The control system switchgear shall be comprised of individual cubicles for each engine generator set plus auxiliary cubicle or cubicles as required and shall be factory assembled to form a station switchboard lineup.

The switchboard lineup shall be free standing, self supporting, NEMA 1 (IP 22) metal enclosed for indoor service with internal steel barriers forming control breaker, bus compartment and hinged front doors for access to control and breaker compartments. Open bottom rear area shall be provided in each cubicle for field cable entrance/exit.

The minimum instrumentation, controls, and protective devices for each engine generator set cubicle, shall be as follows:

1 - AC ammeter, 2% accuracy.

1 - AC voltmeter, 2% accuracy.

1 - Frequency meter, dial type, 45-65 Hz range.

1 - Ammeter phase selector switch.

1 – Voltmeter phase selector switch.

1 - Voltage adjust rheostat, wired to terminal board.

1 - Automatic start-stop cranking module

1 - Mode selector switch for automatic/manual/stop/off-reset operation plus necessary logic circuits for operation with the engine. In the AUTO mode the engine shall be automatically programmed by the system circuits

and the generator set shall automatically be paralleled onto the bus. In the MANUAL mode, the engine shall start and attain operating conditions but the generator set breaker shall not automatically close onto the bus -- to parallel or place the generator set on the bus, it shall be necessary for a station operator to turn the synchronizing switch to its ON position and to use the breaker control switch to close onto the bus. In the STOP position, the generator breaker shall trip open and then engine shall shut down after the cooldown run. In OFF/RESET the controls shall be de - energized and critical engine faults shall be reset. Any critical fault shall also cause tripping of the generator circuit breaker. Engine cranking/fault shutdown controls shall be DC operated of the engine battery system.

4 - Fault indicating lamps, push to test, for shutdown conditions of high water temperature, low oil pressure, overspeed, and overcrank.

1 - Lamp test

1 - Shunt trip, 24 volt Dc, on circuit breaker wired to terminal board.

2 - Auxiliary switches on circuit breaker, one on generator output and one on circuit breaker position, both with one open and one closed set of contracts wired to terminal board.

3 - Current transformers, 5 ampere secondaries.

2 - Potential transformers, 120 volt secondaries, with primary and secondary fuses. Potential transformers shall be provided for voltages above 208 volts.

2 - Ground connector.

1 - Set of control cables, terminal boards, fuses, fuse blocks, nameplates, and miscellaneous hardware as needed, completely installed.

1 - Set of spare fuses and lamps.

1 - Set of software consisting of: wiring diagram, installation dimensions, bill of material (parts list), installation instructions, and operating instructions.

1 - Generator available circuit with indicating lamp to determine when generator set at approximately 90% voltage,

single phase sensing and field adjustable -- circuit provides portion of close signal to the generator circuit breaker in conjunction with the automatic synchronizing system.

1 - Set of auxiliary contacts, two normally open / normally closed (form C), that function after generator set terminates cranking and wired to terminal board points for remote manufacturer.

1 - Mounting and wiring of the engine automatic load share parallel governor module furnished by the engine manufacturer.

1 - Manual speed adjust potentiometer, 10 turns type, for use with the automatic load share governor module.

1 – PLC and I/Os.

1 - Automatic synchronizing circuits to sense and compare the incoming engine generator set voltage, frequency, and phase angle with the bus and to provide automatic close signal to the incoming generator set circuit breaker when voltage, frequency and phase angle of both sources are within synchronizing tolerances. Synchronizing monitor shall control speed and phase angle for the incoming engine generator set and provide automatic correction signals for frequency (speed) differences to the engine automatic load share governor module.

1 - Control circuit to prevent more than one engine generator set breaker from simultaneously closing to a dead bus.

1 - Mounting and wiring of the generator automatic voltage regulator devices -- if regulator devices are not an integral part of and/or located in the generator.

1 - Voltage adjust rheostat for use with the automatic voltage regulator devices.

1 - Breaker control transformer, fused protected, fixed mounted and to provide generator AC potential to the generator breaker operating mechanism.

1 - Generator breaker control switch with positions trip/lock-out, auto close plus position indicating lamps.

1 - Synchronizing switch of the removable key handle type.

1 - Fail to automatically parallel fault lamp circuit and when activated, shall cause engine shutdown.

1 - Control Circuit to prevent auto - reclose of generator breaker should breaker trip due to overcurrent fault plus visual indication of fault -- shall cause engine shutdown.

1 - Lamp to illuminate when engine cranking.

1 - Lamp test pushbutton.

1 - Genset emergency stop pushbutton, pull or twist to reset type shall also cause tripping of generator main circuit breaker.

1 - Set of control cables, terminal boards, fuses, fuseblocks, nameplates, and miscellaneous hardware as needed, completely installed.

1 - Set of spare fuses and lamps.

1 - Set of software consisting of: wiring diagram, installation dimensions, bill of material (parts list), installation instructions, and operating instructions.

## PART 4 - EXECUTION

### 4.1 SHOP INSPECTION AND TESTS

A. The engine-generator set shall be completely assembled in the manufacturer's plant and tested. For the purpose of these tests the engine-generator sets shall be assembled with controls and accessories necessary to demonstrate performance and functioning similar to those in service. Auxiliaries that do not vitally affect the unit or system performance may be other than those intended for use at the site of final installation. The units shall be run in accordance with the manufacturer's standard procedure after which tests shall be made for the following items:

1. Load test

    a. No load - 5 mins

    b. 25% load - 30 mins

    c. 50% load - 30 mins

d. 75% load - 30 mins

e. 100% load - 1 hour

During above test the following data shall be recorded

(1) Line to line voltage
(2) Line current
(3) Frequency
(4) Lub. oil pressure
(5) Lub. oil temperature
(6) Water temperature
(7) Engine speed
(8) Fuel consumption

2. Protection test

a. Low oil pressure

b. High water temperature

c. Overspeed

d. Overcurrent

e. Undervoltage

f. Overvoltage

g. Reverse power

3. Instrument check

a. Voltmeter

b. Ammeter

c. Kw meter

d. Power factor meter

e. Synchroscope

f. Frequency meter

4. Operational Check

a. Manual mode

(1) Start
(2) Stop
(3) Paralleling

b. Auto mode

(1) Start on normal power fail
(2) Auto paralleling
(3) Load sharing
(4) Load monitoring
(5) Stop on normal power return

c. Test mode

(1) Start
(2) Auto paralleling
(3) Stop

B. The test schedule including the date, site and test item shall be submitted to the Owner at least one (1) month in advance. The Owner's engineer will be at the site to witness the tests. After completion of the tests, four (4) copies of suitable bound certified test reports shall be submitted to the Owner for approval within thirty(30) days.

## 4.2 FIELD TESTS AND INSPECTION

1. The unit shall be inspected for correct installation and tested for operation and compliance with the specifications. Tests shall include manual starting, manual synchronizing, manual load sharing, manual stopping, automatic starting, automatic supervision, auto synchronizing, active load sharing, reactive load sharing, load dependent operation and automatic stopping.

2. All defects disclosed by the tests shall be rectified and the tests repeated.

3. An inspection check list shall be prepared for the unit and the auxiliary equipment. An operational test shall be performed to ensure proper operation of all systems.

4. The fuel used in the field tests shall be the same as the fuel to be used in operating the plant.

## 4.3 MANUFACTURER'S REPRESENTATIVE

The services of a manufacturer's representative shall be provided.

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